

PPTStab User Manual

Predicting and Designing Thermostable Proteins

1. Introduction

PPTStab is a tool designed to predict and design thermostable proteins based on their melting temperature (T_m). It uses an ensemble regressor model combining an Artificial Neural Network (ANN) and a Multi-Layer Perceptron (MLP). Both a standalone command-line version and a web-server are available.

For detailed information including the underlying algorithms, visit the PPTStab web server:

<https://webs.iitd.edu.in/raghava/pptstab>

2. Installation

2.1 Install using environment.yml

Create a Conda environment with a name of your choice using the -n flag:

```
conda env create -n <env_name> -f environment.yml
```

Example using the name pptstab:

```
conda env create -n pptstab -f environment.yml
```

Note: If an environment with that name already exists, remove it first: `conda env remove -n <env_name>`

Then activate the environment:

```
conda activate <env_name>
```

2.2 Required Dependencies

The standalone version requires Python 3.11 and the following libraries:

Package	Version
numpy	1.26.4
scikit-learn	1.6.1
tensorflow	2.16.1
torch	2.3.1
transformers	4.41.2
pandas	2.3.3
onnxruntime	1.18.0
joblib	1.4.2
tqdm	4.66.4

3. Usage

3.1 Minimum Usage

To view all available options:

```
python pptstab.py -h
```

To run a basic prediction on the example file (lysate context, default SER method):

```
python pptstab.py -i example.fasta -f 1
```

Results will be saved to outfile.csv by default.

3.2 Full Usage

```
python pptstab.py [-h] [-i INPUT] [-o OUTPUT] [-j {1,2}] [-d {1,2}] [-f {0,1}] [-m {EMB,AAC,SER}]
```

3.3 Command-Line Arguments

Argument	Description	Default
-i INPUT	Input protein/peptide sequences in FASTA format or one sequence per line	Required
-o OUTPUT	Output CSV file name	outfile.csv
-j {1,2}	Job type: 1 = Predict, 2 = Design	1
-f {0,1}	Flag: 1 = lysate (cell-free), 0 = cell	1
-d {1,2}	Display: 1 = thermophilic only, 2 = all proteins	1
-m {EMB,AAC,SER}	Method: EMB (ProtBERT), AAC (amino acid composition), SER (Shannon entropy)	SER

Flag note: The flag is named 'Cell Flag' but value 1 = lysate (cell-free system) and 0 = cell. The default (1) corresponds to lysate context.

3.4 Usage Examples

Predict Tm with default SER method, lysate context:

```
python pptstab.py -i example.fasta -f 1
```

Predict Tm with AAC method, cell context, show all results:

```
python pptstab.py -i example.fasta -f 0 -m AAC -d 2
```

Predict Tm using ProtBERT embeddings (GPU recommended):

```
python pptstab.py -i example.fasta -f 1 -m EMB
```

Design thermostable mutants with SER method:

```
python pptstab.py -i example.fasta -j 2 -f 1 -m SER -o design_output.csv
```

Save predictions to a custom output file:

```
python pptstab.py -i example.fasta -f 1 -o my_results.csv
```